Cpts 327 Quiz 3

Instructions

- **Format Requirement:** Please round (nor floor or ceil) all numerical answers to **two decimal places** (Please use fractions in intermediate steps to avoid precision loss and only round at the very end), e.g., round the result 4.555 to 4.56, 4.444 to 4.44. If your calculated answer has fewer than two decimal places (e.g., 0.1), you must pad it with a trailing zero (e.g., enter **0.10**) to ensure the grading system accepts it.

Problem 1: (10 Points) Let $G : \{0,1\}^n \to \{0,1\}^{\ell(n)}$ be a secure PRG with fixed polynomial $\ell(n) > n$. Alice constructed a $G^{**} : (\{0,1\}^n)^m \to (\{0,1\}^{\ell(n)})^m$ for $x_1, x_2, \ldots, x_m$ where $x_i \in \{0,1\}^n$, the output is defined as:

$$G^{**}(x_1, \ldots, x_m) := (G(x_1 \oplus 1^n) \parallel G(x_2 \oplus 1^n) \parallel \cdots \parallel G(x_m \oplus 1^n)) \oplus 1^{\ell(n)m}.$$

Is G^{**} a secure PRG?

- **A. Yes.** The proof is by reduction, i.e., assume there exists an adversary $\mathcal{A}$ distinguishes the output of G^{**} from a random string, we can build a distinguisher $\mathcal{B}$ for G that embeds its challenge z into a random slot i by computing $z \oplus 1^{\ell(n)}$ as the i-th block, while simulating the other $m-1$ blocks using G and independent seeds to break G via a hybrid argument.
- **B. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{D}$ can distinguish a single instance of the original generator G, we can construct an attack on G^{**} by XORing the challenge with $1^{\ell(n)}$ and padding the remaining $m-1$ slots with random bits to simulate a full output.
- **C. No.** The construction violates the definition.
- **D. No.** The size of the effective seed space allows for a distinguisher with non-negligible advantage in polynomial time.

Problem 2: (10 Points) Let $F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ be a secure PRF. We define a new function $F' : \{0,1\}^n \times (\{0,1\}^n \times \{0,1\}^n) \to \{0,1\}^n$ as:

$$F'(k, (x, y)) = F(k, x) \oplus F(k, y)$$

The PRF Distinguishing Game:

- **World 0 ($b = 0$):** The challenger samples a uniform key $k \leftarrow \{0,1\}^n$ and computes $z_i = F'(k, (x_i, y_i))$ for each queried pair.
- **World 1 ($b = 1$):** The challenger samples a truly random function $f \leftarrow$ Funs$[\{0,1\}^n, \{0,1\}^n \times \{0,1\}^n]$ and computes $z_i = f(x_i, y_i)$ for each queried pair.

Alice proposes an adversary $\mathcal{A}_{Alice}$ to interact with the challenger. For $n = 10$, her strategy is as follows:

(a) Alice defines two complementary base strings: $u = 0^{10}$ and $v = 1^{10}$.

(b) She constructs 10 distinct input pairs $P_0, P_1, \ldots, P_9$ by flipping bits from left to right:

$$P_i = (u \oplus 0^{10-i}1^i,\ v \oplus 0^{10-i}1^i)$$

(c) She queries the challenger for the set $\{P_0, \ldots, P_9\}$ and receives $\{z_0, \ldots, z_9\}$.

(d) If she finds any i, j such that $z_i = z_j$ (for $i \neq j$), she outputs $b' = 0$; otherwise, she outputs $b' = 1$.

Questions:

(a) Calculate $P_1 = \Pr[b' = 0 \mid b = 1]$, the probability that Alice outputs 0 in World 1.

(b) Calculate $P_0 = \Pr[b' = 0 \mid b = 0]$, the probability that Alice outputs 0 in World 0.

(c) Calculate Alice's advantage $Adv = |P_1 - P_0|$.

(d) Do you think F' is a secure PRF?